

YASH DESHPANDE

Senior Scientist | Machine Learning for Omics and Target Discovery

☎ (+1) 917 362 9676 ✉ yashrajas@gmail.com in [linkedin.com/in/yash-deshpande](https://www.linkedin.com/in/yash-deshpande) 🐙 github.com/yashd94

SUMMARY

Machine learning scientist with experience developing computational methods and scalable pipelines for single-cell genomics, statistical genetics, multi-modal data integration, companion diagnostic development, and translational biomarker discovery in biopharma. Currently developing single-cell RNA-seq models for genetic perturbation response and histopathology cell type segmentation models for target prioritization. Strong cross-functional leadership through close collaboration with translational scientists, computational biologists, and domain stakeholders.

SKILLS

Data: Single-cell & bulk RNA-seq, GWAS, single-cell genetic perturbation screens, H&E and IHC histopathology slides

Programming: Python (pandas; numpy; scikit-learn; PyTorch; Tensorflow), R, SQL, Shell

Infrastructure: Nextflow, AWS, Docker, Git, MLFlow, W&B

PROFESSIONAL EXPERIENCE

Merck & Co. | Senior Scientist, Machine Learning

Dec 2023 - Present

- Develop machine learning models for genetic perturbation response using single-cell RNA-seq data to support target discovery and prioritization, linking perturbation-driven transcriptional programs to candidate biological mechanisms
- Develop cell segmentation models using HE-stained histopathology data to quantify cell types of interest and derive imaging biomarkers for downstream translational and companion diagnostics applications.
- Apply ML and statistical methods to high-dimensional molecular datasets with an emphasis on interpretable, robust, and reproducible biological insights
- Collaborate with computational biologists and translational research teams to align perturbation modeling outputs with target discovery and mechanistic interpretation objectives

Prometheus Biosciences | Machine Learning Engineer

Jun 2022 - Dec 2023

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- Led development of Minerva, a multi-modal companion diagnostics (CDx) discovery platform within Prometheus360, enabling versioned, reproducible, and extendable model development using multi-modal -omics data
- Built scalable Nextflow pipelines for whole-genome scanning, QTL analysis, and evidence meta-analysis to prioritize SNP features for downstream CDx modeling and machine learning
- Developed and built pipelines for genotype clustering and Bayesian classification modeling workflows for CDx development downstream of SNP discovery
- Collaborated closely with immunologists, computational biologists, radiologists, and -omics experts to align multi-modal methods and platform development with translational priorities, biomarker strategy, and CDx discovery

Fresenius Medical Care North America | Machine Learning Engineer

Oct 2020 - May 2022

- Built cloud-based capabilities for an internal AI workbench on AWS, enabling scalable model monitoring and faster analytical iteration across cross-functional teams
- Developed reusable ML infrastructure and tooling to improve adoption of cloud-native workflows and accelerate time to insight
- Built a Python-based NLP model that captured 40+ previously unreported instances of abdominal pain annually from unstructured clinical text

ZS | Data Analyst, Business Technology Group

Jul 2016 - May 2018

- Reduced front-end processing time by 70% on a client-facing business rules platform through performance-focused workflow improvements
- Developed and maintained ETL pipelines using shell, PostgreSQL, and AWS Redshift to optimize recurring analytics processes

EDUCATION

New York University
Master of Science in Data Science

Sep 2018 - May 2020

University of Pune
Bachelor of Engineering in Electronics & Telecommunication Eng.

Aug 2012 - May 2016

SELECTED PUBLICATIONS

- Gerardo A. Flores et al. *Pandora's Regret: A Proper Scoring Rule for Evaluating Sequential Search*. 2026. arXiv: 2605.01936 [cs.LG]. URL: <https://arxiv.org/abs/2605.01936>
- Ruiwen Ding et al. *Predicting ulcer in HE images of inflammatory bowel disease using domain-knowledge-driven graph neural network*. 2025. arXiv: 2504.09430 [eess.IV]. URL: <https://arxiv.org/abs/2504.09430>
- Houda Alberts et al. "VisualSem: a high-quality knowledge graph for vision and language". In: vol. abs/2008.09150. 2020. arXiv: 2008.09150. URL: <https://arxiv.org/abs/2008.09150>
- Ningyuan Huang et al. "Endowing Language Models with Multimodal Knowledge Graph Representations". 2022. arXiv: 2206.13163 [cs.CL]. URL: <https://arxiv.org/abs/2206.13163>

LANGUAGES

Fluent: English, Marathi, Hindi
Basic: Japanese